



Universidad Modelo
Escuela de Ingeniería

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Ingeniería en Desarrollo de Tecnología y Software

Administración de Bases de Datos

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Práctica 1. Caso de Estudio: Diseño de Base de Datos

Ingresar al contenedor e ingresar a base de datos

```
millie — docker exec -it mysql-container bash — 196x30
Last login: Fri Aug 28 20:01:39 on ttys000
(base) millie@Millies-MacBook-Pro ~ % docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
807dd59089b5   mysql:8        "docker-entrypoint.s..." 3 days ago    Up 14 minutes    0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp    mysql-container
f15b0665374b   hello-world    "/hello"                 3 days ago    Exited (0) 3 days ago                               naughty_blackburn
4b1709bde924   mcr.microsoft.com/mssql/server:2022-latest "/opt/mssql/bin/laun..." 5 months ago  Exited (137) 5 months ago                               mssql_server
5c14095cf15f   mcr.microsoft.com/mssql/server:2022-latest "/opt/mssql/bin/laun..." 7 months ago  Exited (137) 5 months ago                               sqlserver
(base) millie@Millies-MacBook-Pro ~ % docker start mysql-container
mysql-container
(base) millie@Millies-MacBook-Pro ~ % docker exec -it mysql-container bash
bash-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.11 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

Creación de base de datos

```
millie — docker exec -it mysql-container bash — 196x49

mysql> create database practical1;
Query OK, 1 row affected (0.01 sec)

mysql> use practical1;
Database changed
mysql> CREATE TABLE usuarios (
  ->   id_usuario INT PRIMARY KEY,
  ->   nombre_usuario VARCHAR(100),
  ->   email_usuario VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE libros (
  ->   id_libro INT PRIMARY KEY,
  ->   titulo_libro VARCHAR(200),
  ->   autor_libro VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE prestamos (
  ->   id_prestamo INT PRIMARY KEY,
  ->   fecha_prestamo DATE,
  ->   id_usuario INT NOT NULL,
  ->   id_libro INT NOT NULL,
  ->   FOREIGN KEY (id_usuario) REFERENCES usuarios(id_usuario),
  ->   FOREIGN KEY (id_libro) REFERENCES libros(id_libro)
  -> );
Query OK, 0 rows affected (0.02 sec)

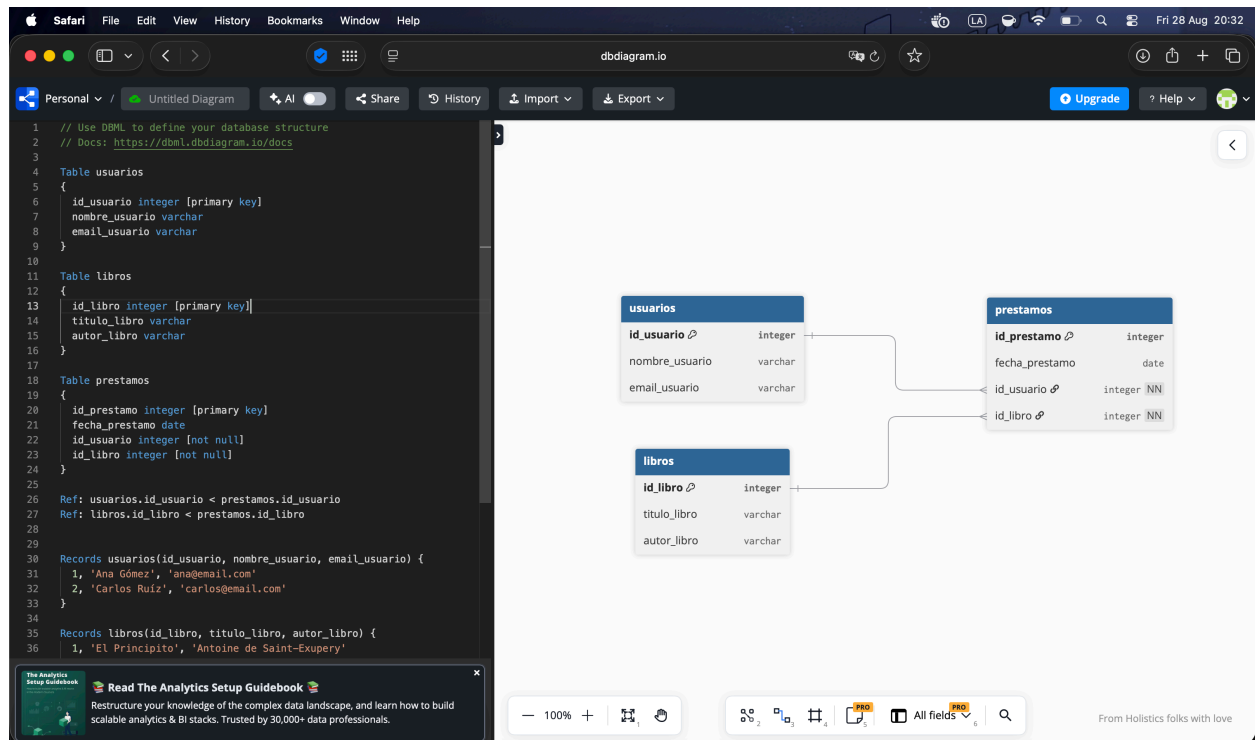
mysql> INSERT INTO usuarios (id_usuario, nombre_usuario, email_usuario) VALUES
  -> (1, 'Ana Gomez', 'ana@email.com'),
  -> (2, 'Carlos Ruz', 'carlos@email.com');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO libros (id_libro, titulo_libro, autor_libro) VALUES
  -> (1, 'El Principito', 'Antoine de Saint-Exupery'),
  -> (2, 'Cien Años de Soledad', 'Gabriel Garcia Marquez');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO prestamos (id_prestamo, fecha_prestamo, id_usuario, id_libro) VALUES
  -> (5001, '2026-03-01', 1, 1),
  -> (5002, '2026-03-02', 2, 2),
  -> (5003, '2026-03-05', 1, 2);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
```

Diagrama



Tablas Creadas

```
mysql> show tables;
+-----+
| Tables_in_practica1 |
+-----+
| libros               |
| prestamos            |
| usuarios              |
+-----+
3 rows in set (0.00 sec)

mysql> select * from usuarios;
+-----+-----+-----+
| id_usuario | nombre_usuario | email_usuario |
+-----+-----+-----+
| 1          | Ana Gmez       | ana@email.com |
| 2          | Carlos Ruz     | carlos@email.com |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> select * from prestamos;
+-----+-----+-----+-----+
| id_prestamo | fecha_prestamo | id_usuario | id_libro |
+-----+-----+-----+-----+
| 5001        | 2026-03-01    | 1          | 1        |
| 5002        | 2026-03-02    | 2          | 2        |
| 5003        | 2026-03-05    | 1          | 2        |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from libros;
+-----+-----+-----+
| id_libro | titulo_libro      | autor_libro      |
+-----+-----+-----+
| 1        | El Principito     | Antoine de Saint-Exupery |
| 2        | Cien Aos de Soledad | Gabriel Garcia Marquez |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```